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TriMNet-P: Tri-Modality Network for Phishing Attack Detection on Social and Digital Communication Platforms

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Abstract

Phishing attacks are increasingly spreading through social media channels using malicious URLs, spoofed websites, and malicious attachments that capture personal information. Conventional detection systems are at a loss when detecting such multimodal attacks because they are based on single-modality analysis. To overcome detection of such attacks, the proposed TriMNet-P (Tri-Modality Network for Phishing Attack Detection) uses deep learning to combine detection of textual, visual, and network behavioural features. A Context-Aware Modality Attention (CAMA) mechanism dynamically weights relevant inputs across email, WhatsApp, and Instagram platforms. Tested on Phishing Email, Open Phish, and CIC-IDS2017 datasets, TriMNet-P has attained 98.6% accuracy, 97.9% precision, 97.5% recall, and runs at only 6ms latency.

Keywords: *Phishing Attack, Social Media, Text Analysis, Image Analysis, Network Analysis, Context-Aware Modality Attention (CAMA), Cybersecurity, Machine Learning*

1. Introduction

Phishing attacks compromise sensitive personal information such as credit card numbers and login credentials by deceiving victims. Phishing attacks have proliferated on social media platforms, where fraudsters impersonate trusted individuals or brands to deceive victims. Traditional detection by text-based analysis like NLP relies on suspect URLs and keywords, effective against email phishing but not against advanced multimodal phishing using images and videos. Modern phishing scams are more likely to use visual deceptions and suspicious network activity, such as invocation of hijacked accounts and suspicious IP addresses. Attackers generally use spoofed sites masquerading as authentic sites, send malicious links via email or social media, and scrape victims' credentials for unauthorized login [1]. Phishing attacks produce indicators such as anomalous IP addresses, spoofed identities, unusual message volumes, timing anomalies, and activity with bot-like accounts. These malicious activity patterns appear innocuous but can be detected by network metadata analysis [2].

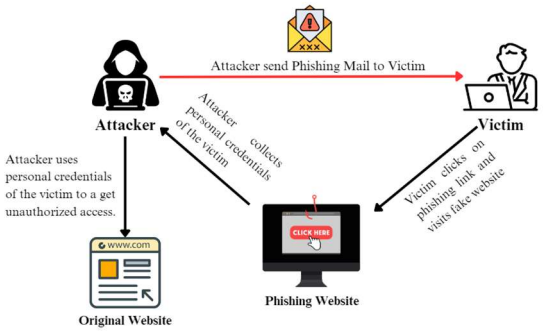

Figure 1: Phishing Attack

Phishing constitutes a serious cybersecurity threat, accounting for over 36% of data breaches, according to the 2023 Verizon report. Attackers employ multimodal features that include text, image, and network behaviours to design sophisticated attacks, which are challenging to defend against using single-modality-based detection systems. This paper, therefore, proposes TriMNet-P, a deep learning model that integrates text (URLs), image (decoy logos), and network behavioural patterns with a Context-Aware Modality Attention (CAMA) mechanism. TriMNet-P dynamically weights each modality's contribution to improve detection performance and is evaluated on combined benchmark datasets with multiple metrics.

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The paper's organization is as follows: Section 2 contains an overview of related work, Section 3 describes the proposed method, Section 4 provides experiments and results, and Section 5 concludes with the key findings and future research directions.

2. Related Work

Rao et al. (2019) [3] compared multiple machine learning models for phishing detection and reported that Random Forest gained 99.31% accuracy on a malicious and legitimate website dataset. Suryan et al. (2020) [4] compared multiple classifiers on a UCI dataset, with Random Forest reporting 96% accuracy with 97% precision. Al-Sarem et al. (2021) [5] proposed an ensemble stacking model using gradient boosting methods and reported 98.58% accuracy on URL-based phishing datasets. Butnaru et al. (2021) [6] reported that Random Forest outperformed SVM, MLP, and Naive Bayes on an imbalanced phishing dataset with 99.29% accuracy. Cuzzocrea et al. (2019) [7] utilized decision tree models and concluded that J48 and RepTree were optimal classifiers for phishing detection on UCI data. Cristian et al. (2020) [8] proposed PhishKiller, a CNN-LSTM model, which reported 98.30% accuracy with 81 ms processing time. Tharani et al. (2020) [9] compared KNN and SVM on Alexa, Phish Tank, and Open Phish datasets with high accuracy using 10 features. Sameen et al. (2020) [10] utilized an SVM ensemble with lexical feature analysis and reported 98% accuracy and 97% precision for URL phishing detection. Alsariera et al. (2020) [11] utilized ensemble models such as bagging and Logit Boost and reported 97% accuracy on the UCI phishing dataset. Naaz et al. (2021) [12] evaluated RF, DT, Neural Networks, and SVM on the UNSW-NB15 IoT dataset, with Random Forest achieving 96% accuracy. Sirinavasa et al. (2022) [13] developed a multimodal ensemble model that reached 99.34% accuracy with low false positive rates in textual and domain-specific phishing detection. Almalaq et al. (2022) [14] combined Random Forest with AdaBoost in a multi-grid system, obtaining 93.91% accuracy, and explored deep learning enhancements. Chen et al. (2021) [15] proposed a reinforcement learning and blockchain-based model to secure smart grids, demonstrating strong phishing detection accuracy.

Despite advances in phishing detection, current methods face limitations, including reliance on unimodal analysis, static modality weighting, limited datasets, lack of detailed feature selection, and insufficient focus on real-time adaptability. Key research gaps include:

- Reliance on text-only detection methods limits multimodal attack coverage.
- Static weighting of modalities without dynamic context adaptation.
- Use of limited, domain-specific datasets lacking real-world diversity.
- Absence of granular phishing-specific features like logo similarity and network anomalies.
- Inadequate focus on low-latency, real-time detection.
- Testing models in static conditions without evaluating adaptability to evolving attacks.
- Limited consideration for practical deployment challenges.

The proposed approach addresses these issues through dynamic multimodal attention, comprehensive datasets, targeted feature selection, and a low-latency design, with future work aiming to incorporate hybrid models and advanced learning techniques.

3. Proposed Methodology

TriMNet-P is a phishing detection model that integrates text, image, and network behaviour analysis. Its main components include the tri-modality network architecture, the Context-Aware Modality Attention (CAMA) mechanism, multimodal fusion, and the decision-making process. These components work together to identify phishing attempts and dynamically prioritize the most relevant modality for each case.

3.1. Tri-Modality Network Architecture

The proposed model architecture integrates three key modalities: text, images and network behavioural patterns to detect phishing attacks. It comprises three subnetworks: a text-based subnetwork analysing URLs, domain names, and suspicious linguistic cues using recurrent neural networks (RNNs); an image-based subnetwork employing convolutional neural networks (CNNs) to identify visual phishing indicators such as altered logos and fake certifications; and a network/behavioural subnetwork that evaluates metadata like IP addresses, domain reputation, and sender behaviour to detect anomalies and replicate normal network activity through machine learning techniques.

3.2. Dataset Description & Pre-processing

The TriMNet-P model utilizes three datasets covering text, image, and network modalities. Text data from the Phishing Email dataset [16] (around 20,000 labelled samples) undergoes preprocessing, including tokenization (nltk.word_tokenize or spacy), stop word removal (nltk.corpus.stopwords), and vectorization using TF-IDF (sklearn.feature_extraction.text.TfidfVectorizer) and Word2Vec embeddings (via gensim.models.Word2Vec) to balance efficiency and interpretability. Image data from the Open Phish dataset [17] (over 10,000 labelled images)

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are pre-processed by resizing (e.g., using PIL.Image.resize or cv2.resize) to 224×224 pixels and normalization (scaling pixel values to [0, 1]) typically using libraries like torchvision.transforms.Normalize. Network data from the CIC-IDS2017 dataset [18] includes traffic metadata and behavioural features, which are cleaned and filtered using pandas.DataFrame.dropna(),pandas.DataFrame.fillna() and feature scaling methods such as Min-Max scaling (sklearn.preprocessing.MinMaxScaler). To address class imbalance, stratified sampling (sklearn.model_selection.StratifiedShuffleSplit) was applied to maintain balanced class distributions during training and testing splits. Additionally, oversampling techniques like imblearn.over_sampling.SMOTE were employed to augment minority class samples, improving model generalization synthetically. While transformer-based models like BERT offer advanced text representation, TF-IDF and Word2Vec were chosen here for their computational efficiency and real-time suitability. Future work will explore integrating transformer embeddings to improve performance without compromising latency.

3.3. Relevant Feature Selection

Feature selection was conducted using Random Forest to rank features by importance for phishing detection. The Random Forest Classifier from scikit-learn was trained on the merged dataset, and feature importances were extracted via the feature_importances_ attribute. The top 15 features, shown in Fig. 2, include key phishing indicators from text, image, and network modalities, improving model accuracy by focusing on the most relevant signals.

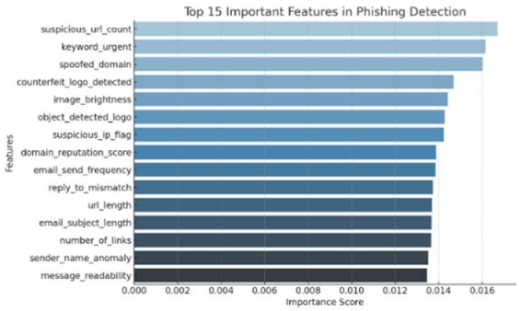

Figure 2: Important 15 Features for Phishing Detection

Figure 3 demonstrates the top 15 features' correlation matrix consisting of three modalities (text, image & network). Then the dataset is divided into a 70:30 ratio for the final evaluation of the proposed hybrid model using various measurement metrics.

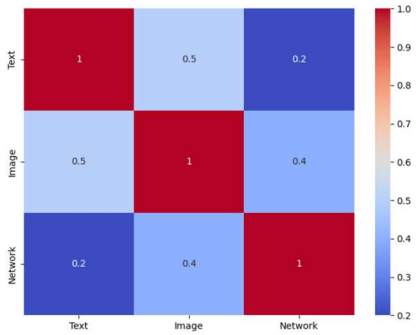

Figure 3: Correlation matrix of the Multimodal features

3.4. Context-Aware Modality Attention (CAMA)

The CAMA mechanism dynamically weights each modality based on contextual relevance, allowing adaptive prioritization for different phishing attacks. Though simple, it offers an interpretable, low-overhead solution for real-time detection. Late-stage fusion with XGBoost enables modular training and flexible integration, balancing performance, scalability, and maintainability. Attention weights for each modality (W_t, W_i, W_n) are computed as follows in (1):

$$W_k = \frac{e^{fk}}{\sum_{j\in\{t,i,n\}} e^{fj}} \tag{1}$$

Where fk Represents the relevance score of modality k. The final phishing probability (P) is calculated as (2):

$$P = W_t \,.\, M_t + W_i \,.\, M_i + W_n \,.\, M_n \tag{2}$$

The Context-Aware Modality Attention (CAMA) mechanism proportionally balances the output of each modality to prioritize different phishing instances, enhancing detection rates. The model combines the weighted output of the text, image, and network subnetworks with a weighted sum: fused_output = sum(w * output for w, output in zip(cama_weights, modality_outputs)). A threshold is applied on the fused output (is_phishing = fused_output >= threshold) to mark communications as phishing or non-phishing. The threshold may be tuned to balance false positive and false negative rates depending on the specific requirements of the detection system.

3.5. Flow of the Proposed Model

The TriMNet-P model, as shown in Figure 4, processes modality data of text, image, and network, achieving significant feature extraction with Random Forest scoring. The text subnetwork uses RNNs, the image subnetwork uses CNNs, and the network subnetwork uses Random Forest for classification. Their outputs are

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merged and classified by XGBoost for final phishing detection.

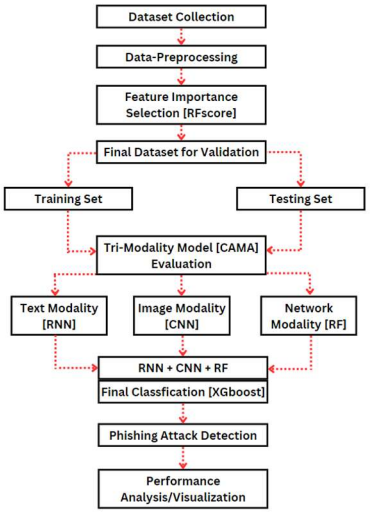

Figure 4: TriMNet-P Hybrid model

4. Results & Analysis

The TriMNet-P model classifies phishing using RNN for text, CNN for images, and Random Forest for network data. An XGBoost classifier fuses these results for final, accurate detection. The model shows strong accuracy, precision, recall, and real-time capability. The following sections detail the experimental setup, dataset, and feature analysis, and they compare TriMNet-P with unimodal and multimodal baselines.

4.1. Evaluation Metrics

The proposed model's performance is evaluated using accuracy, precision, recall, F1-score, latency, and cross-validation. Accuracy measures the correct classification rate, while precision minimizes false positives, which is crucial in practical settings. Recall assesses the model's ability to detect actual phishing cases—F1-score balances precision and recall, which is essential for imbalanced datasets. Latency measures the processing time, and cross-validation evaluates the model's consistency.

4.2. Performance Analysis

Figure 5 shows the TriMNet-P model's performance, achieving 98.6% accuracy and 97.9% precision in classifying phishing attacks across three modalities. The high precision indicates effective reduction of false positives, making the model well-suited for real-world applications.

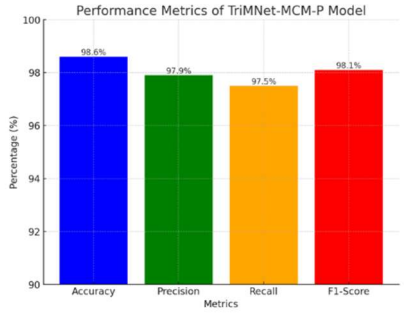

Figure 5: Performance Metrics Results

Figure 6 showcases the cross-validation results of the TriMNet-P model, which demonstrates consistent performance in accuracy across all 10 folds, with an accuracy rate between 98.0% and 98.6%. It proves that the model is working consistently with high performance in detecting phishing attacks.

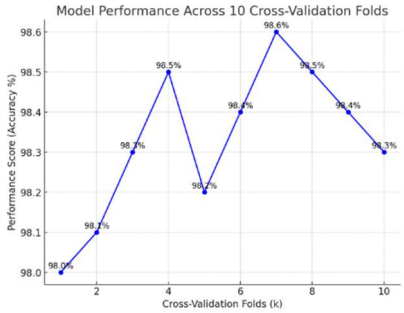

Figure 6: Cross-Validation Results

The TriMNet-P model's latency ranges from 6.22 to 6.6 ms across 10-fold cross-validation as shown in Figure 7, demonstrating consistent, efficient performance suitable for real-time phishing detection in applications like social media monitoring and email filtering.

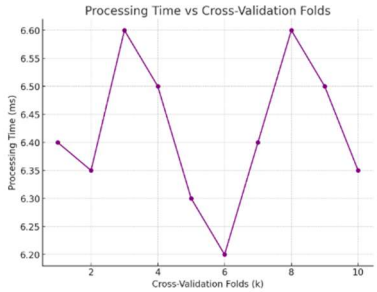

Figure 7: Latency (Processing Time) Results

Figure 8 presents the ROC curve of the TriMNet-P model, showing a high Area Under the Curve (AUC) of nearly 1, indicating a strong ability to distinguish phishing from legitimate attempts. The model effectively classifies phishing using text, image, and network modalities with

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dynamic weighting via CAMA. Its low latency, high accuracy, precision, recall, and F1-score across 10-fold cross-validation make it suitable for real-time applications.

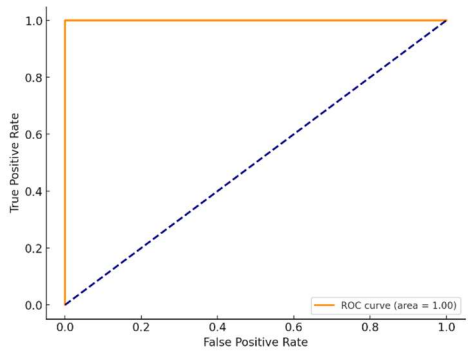

Figure 8: ROC Curve

4.3. Ablation Study on CAMA

Ablation studies showed that the CAMA mechanism improved accuracy by 0.8% and F1-score by 1.5% compared to equal weighting. Removing text caused a significant accuracy drop, while excluding image and network modalities reduced performance moderately and mainly affected recall, respectively. These results confirm CAMA's effectiveness and the importance of all three modalities as shown in Table 1.

Table 1. Ablation Study on CAMA

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Full TriMNet-P with CAMA	98.6	97.9	97.5	98.1
TriMNet-P without CAMA (uniform weights)	97.8	97.0	96.2	96.6
TriMNet-P without Text Modality	94.2	92.5	91.8	92.1
TriMNet-P without Image Modality	95.1	94.0	93.7	93.8
TriMNet-P without Network Modality	93.6	92.1	91.0	91.5

4. Comparative Analysis

Many research studies have employed machine learning algorithms to develop models for phishing attack classification. Table 2 makes a comparative analysis between the existing top-tier models and the proposed hybrid TriMNet-P model to justify the performance enhancements achieved by merging text, image, and network modalities for phishing classification. TriMNet-P achieved 98.6% accuracy with strong metrics, outperforming existing models and showing consistent 10-fold cross-validation results. It integrates RNN, CNN, Random Forest, and XGBoost modularly to leverage their strengths and enable parallel processing. With optimizations, it achieves low latency (~6 ms) suitable for real-time detection. Future work will improve efficiency and explore end-to-end deep learning.

Table 2. Model Comparison with Existing Models

Models	Algorithms	Modality	Dataset	Accuracy
Kamble & Mishra (2024) [19]	FDHPO-SqueezeNet	Text, Image	Own Custom Dataset	93.05 %
Geetha et al. (2024) [20]	CNN, RNN	Text (Phishing Emails)	Own Custom Dataset	94.5 %
Suryan et al. (2020)	Random Forest (RF)	Text (URL-based features)	UCI Machine Learning Repository	96.65 %
Naaz et al. (2021)	Random Forest (RF), SVM	Network (IoT Phishing Detection)	UNSW-NB15 IoT Dataset	96.85 %
Tharani et al. (2020)	SVM, KNN	Text (URLs)	Alexa & PhishTank	90.34%
TriMNet-P (Proposed Hybrid Model)	RNN, CNN, Random Forest, XGBoost	Text, Image, Network	Phishing Email, Open Phish, CIC-IDS2017	98.60 %

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5. Conclusion & Future Scope

The TriMNet-P model integrates text, image, and network behaviour using RNN, CNN, RF, and XGBoost, achieving 98.6% accuracy, 97.9% precision, 97.5% recall, and 98.1% F1-score. It leverages Context-Aware Modality Attention (CAMA) to prioritize relevant modalities for phishing detection, making it suitable for social media, email filtering, and digital communication platforms. Future work includes real-time data expansion to cover mobile scams, vishing, and other cyber threats. TriMNet-P offers a robust foundation against evolving phishing attacks.

Conflict of Interest

The authors declare no conflict of interest

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